

İZMİR UNIVERSITY OF ECONOMICS



Kaoutar Ammara –20210615207

Sena Güven – 20220615019

Sıla Arslan-20220615001

Aerospace Engineering

AE-301 AERODYNAMICS

Homework 3

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ANSYS Simulation and Flow Analysis for a pipe

1. Introduction

This study focuses on the modeling and analysis of incompressible flow (water) within a T-junction pipe network featuring two inlets and a single outlet. The numerical simulations were conducted using ANSYS Fluent to evaluate the outlet velocity and the pressure distribution.

2. Methodology

2.1 Geometric Configuration and Boundary Conditions

The pipe system is characterized by the following features:

- Inlet 1 (Smaller Circular Section): Specified inlet velocity = 20 m/s.
- Inlet 2 (Lower Large Circular Section): Specified inlet velocity = 40 m/s.
- Outlet (Upper Large Circular Section): Specified as a pressure outlet with a gauge pressure of 0 Pa.

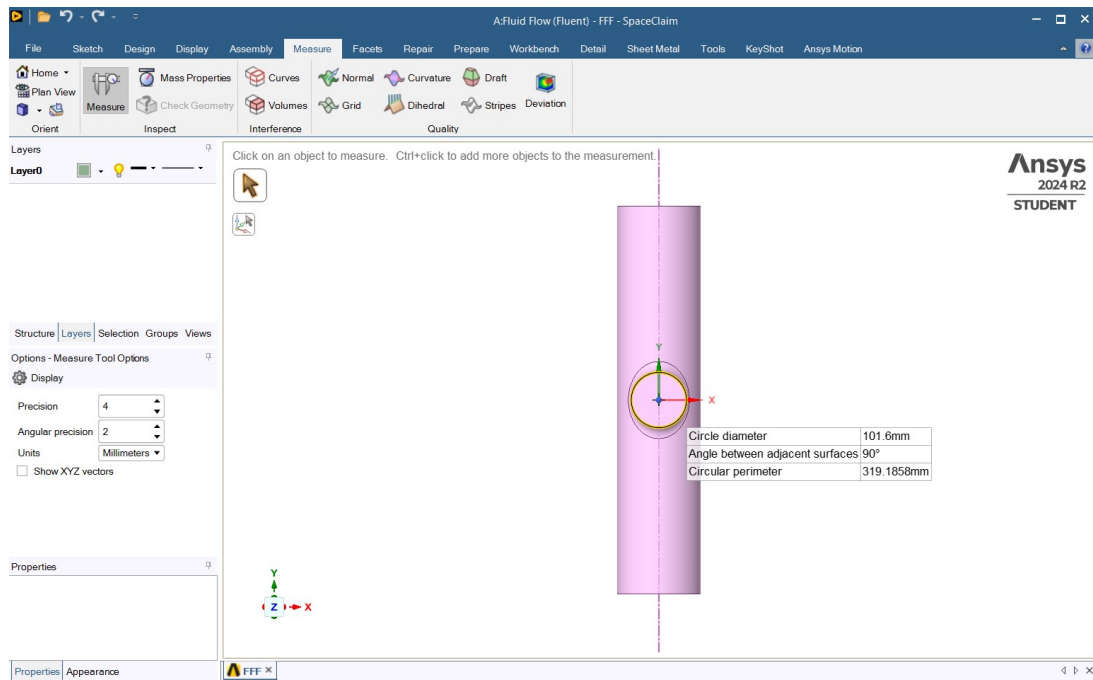
The cross-sectional areas of the pipe sections are determined using the geometric formula for circular cross-sections:

$$A = \pi \times \frac{d^2}{4}$$

The dimensions and corresponding areas (measured from SpaceClaim in Geometry) are as follows:

- Inlet 1 diameter

$$d_1 = 101.6 \text{ mm} = 0.1016 \text{ m}$$

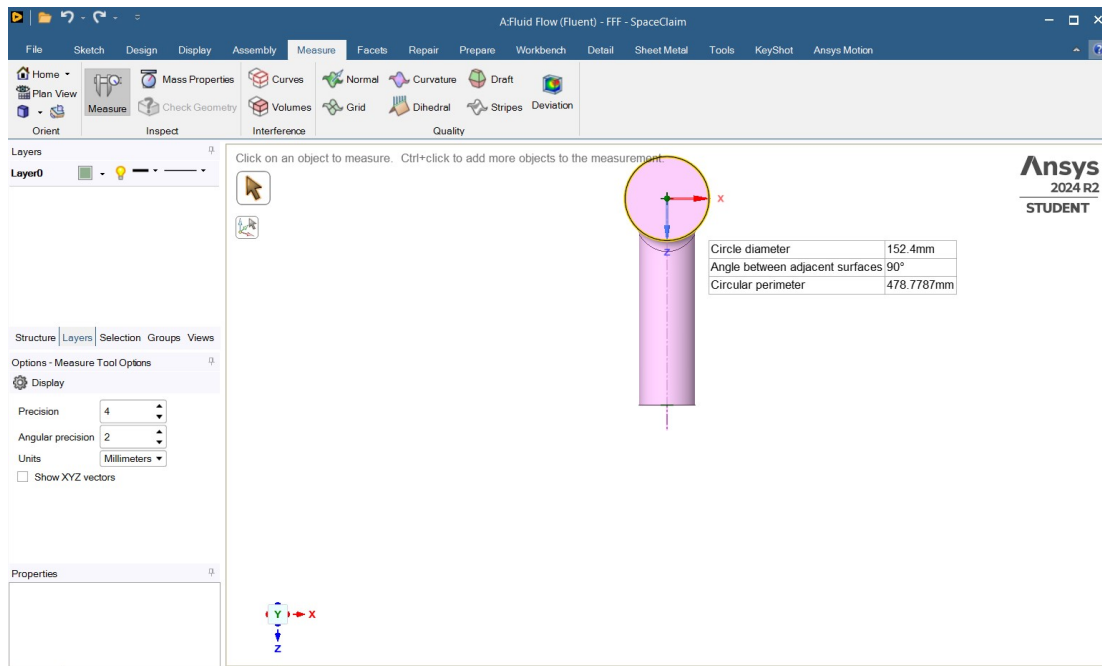


- Inlet 2 diameter:

$$d_2 = 152.4 \text{ mm} = 0.1524 \text{ m}$$

- Outlet diameter:

$$d_{outlet} = 152.4 \text{ mm} = 0.1524 \text{ m}$$



2.2 Theoretical Calculation of Outlet Velocity

The theoretical outlet velocity, v_{outlet} was determined using the principle of mass conservation, expressed as:

$$A_1 v_1 + A_2 v_2 = A_{outlet} v_{outlet}$$

Rearranging for v_{outlet} :

$$v_{outlet} = \frac{A_1 v_1 + A_2 v_2}{A_{outlet}}$$

Substituting the given values:

$$v_{outlet} = \frac{\pi \times \frac{0.1016^2}{4} \times (20 \text{ m/s}) + \pi \times \frac{0.1524^2}{4} \times (40 \text{ m/s})}{\pi \times \frac{0.1524^2}{4}}$$

$$v_{outlet} = 48.888889 \text{ m/s}$$

2.3 Numerical Simulation Setup

The simulation was performed in ANSYS Fluent with the following configurations:

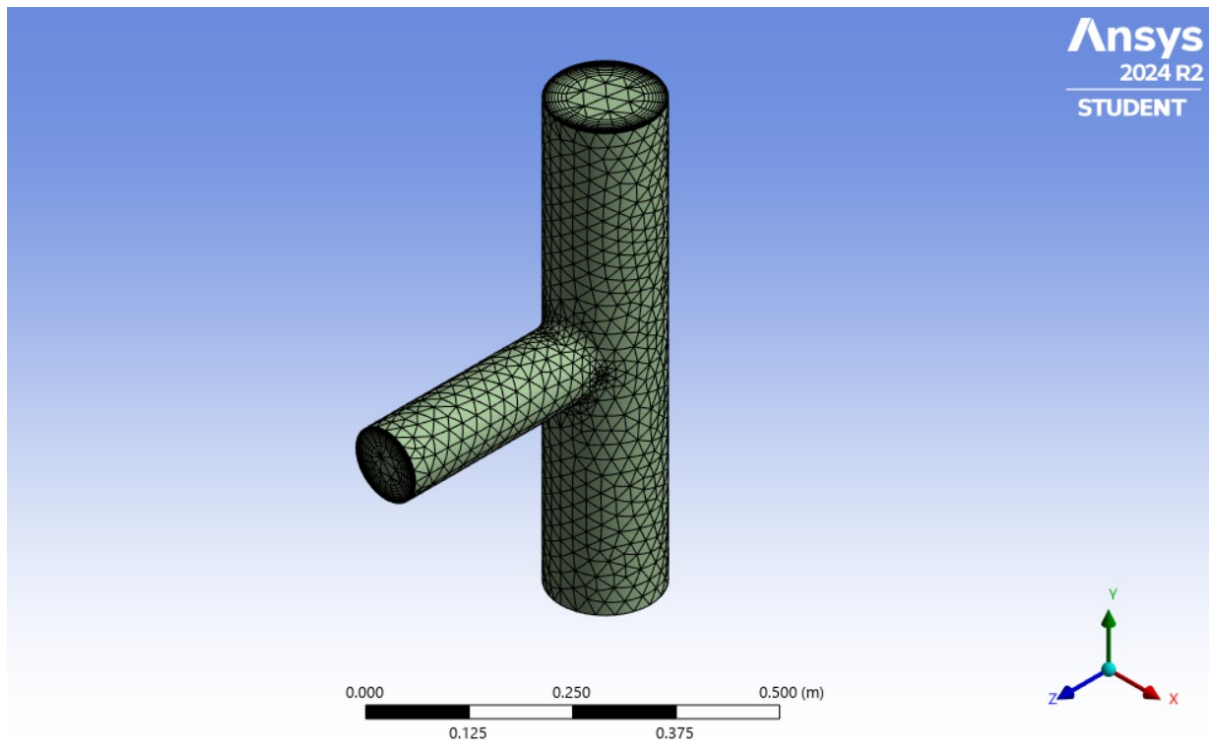
- Turbulence Model: The k- ω SST model, employing Tabulated Wall Treatment for improved near-wall resolution.
- Spatial Discretization: Second-order upwind schemes applied to all governing equations for enhanced accuracy.
- Under-Relaxation Factors (URFs):
 - Momentum = 0.25.
 - Turbulent Kinetic Energy (κ) = 0.5.
 - Specific Dissipation Rate (ω) = 0.5.
- Mesh Details:

Object Name	<i>Mesh</i>
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	

Physics Preference	CFD
Solver Preference	Fluent
Element Size	2.5e-002 m
Export Format	Standard
Export Preview Surface Mesh	No
Sizing	
Use Adaptive Sizing	Yes
Resolution	2
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Slow
Span Angle Center	Fine
Initial Size Seed	Assembly
Bounding Box Diagonal	0.85574 m
Average Surface Area	5.901e-002 m ²
Minimum Edge Length	1.4874e-002 m
Quality	
Check Mesh Quality	Mesh Quality Worksheet
Target Skewness	Default (0.9)
Smoothing	High
Mesh Metric	Skewness
Min	3.667e-003
Max	0.77788
Average	0.2575
Standard Deviation	0.13763
Inflation	
Use Automatic Inflation	Program Controlled
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	10
Growth Rate	1.2
Inflation Algorithm	Pre

View Advanced Options	Yes
Collision Avoidance	Layer Compression
Fix First Layer	No
Gap Factor	0.5
Maximum Height over Base	1
Growth Rate Type	Geometric
Maximum Angle	140.0°
Fillet Ratio	1
Use Post Smoothing	Yes
Smoothing Iterations	5
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Statistics	
Nodes	73663
Elements	32398
Show Detailed Statistics	No

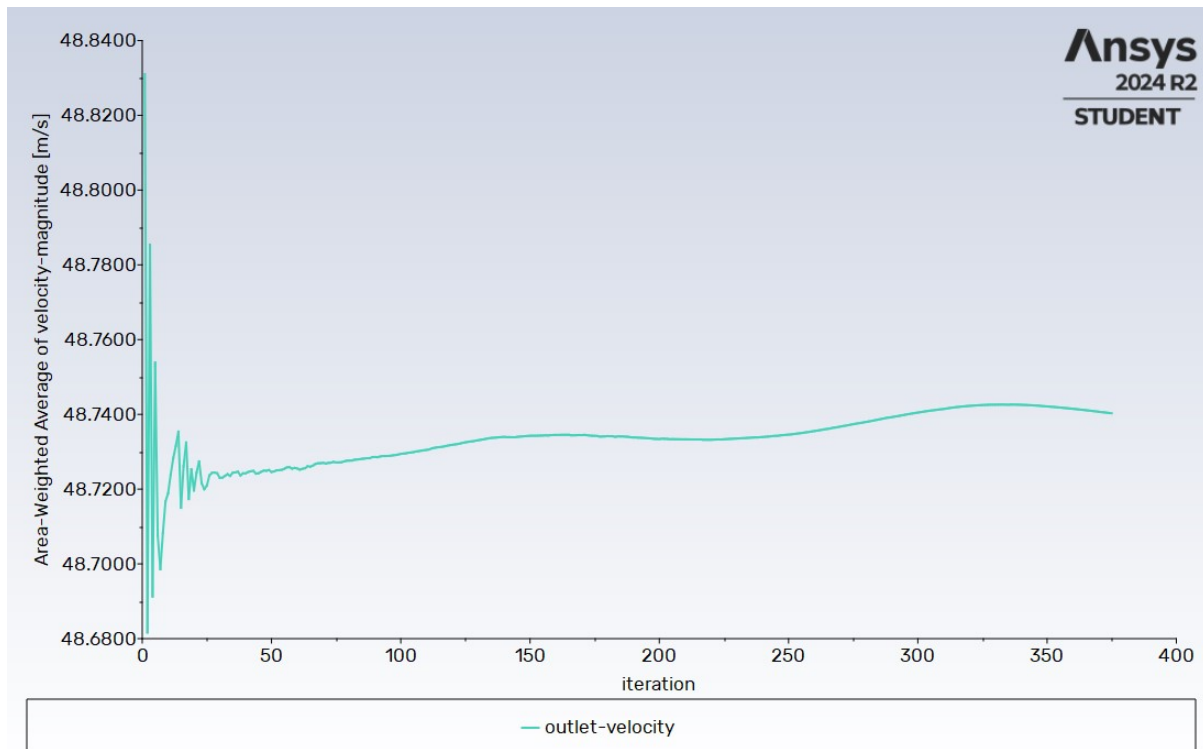
A tetrahedral mesh method was set.



The simulation was iterated for 500 steps to ensure adequate convergence of residuals and stabilization of physical parameters.

2.4 Convergence Criteria

- Residuals for continuity and momentum equations were monitored and reduced below 10^{-3}
- Outlet velocity was tracked to be evaluated at the convergence.



-The solution converged after 375 iterations.

```

iter  continuity  x-velocity  y-velocity  z-velocity          k      omega      time/iter
 375   9.9681e-04  6.3784e-06  1.3924e-05  8.8490e-06  4.6781e-04  5.6328e-06  0:01:47  225
! 375 solution is converged
Registering ReportDefFiles, ("D:\ansys\ppkss_files\dp0\FFF\Fluent\.\report-file-0.out")
Writing "| gzip -2cf > SolutionMonitor.gz"...
Writing temporary file C:\Users\akaou\AppData\Local\Temp\flntgz-142167 ...
Done.

```

The simulation successfully converged within the defined criteria of 10^{-3} . The achieved residuals provide reliable results for engineering analysis. This balance between computational cost and solution accuracy validates the approach used in this study.

3. Results and Discussion

3.1 Simulated Outlet Velocity

The outlet velocity was calculated using the area-weighted average velocity method, ensuring accurate integration of velocity components over the outlet boundary. The simulated outlet velocity obtained was: 48.740287 m/s

```

Done.
Calculation complete.

Area-Weighted Average [m/s]
-----
outlet 48.740287

```

3.2 Error Analysis

The relative error between the theoretical and simulated outlet velocities was computed as:

$$Error = \frac{|Simulated Value - Experimental Value|}{Simulated Value} \times 100$$

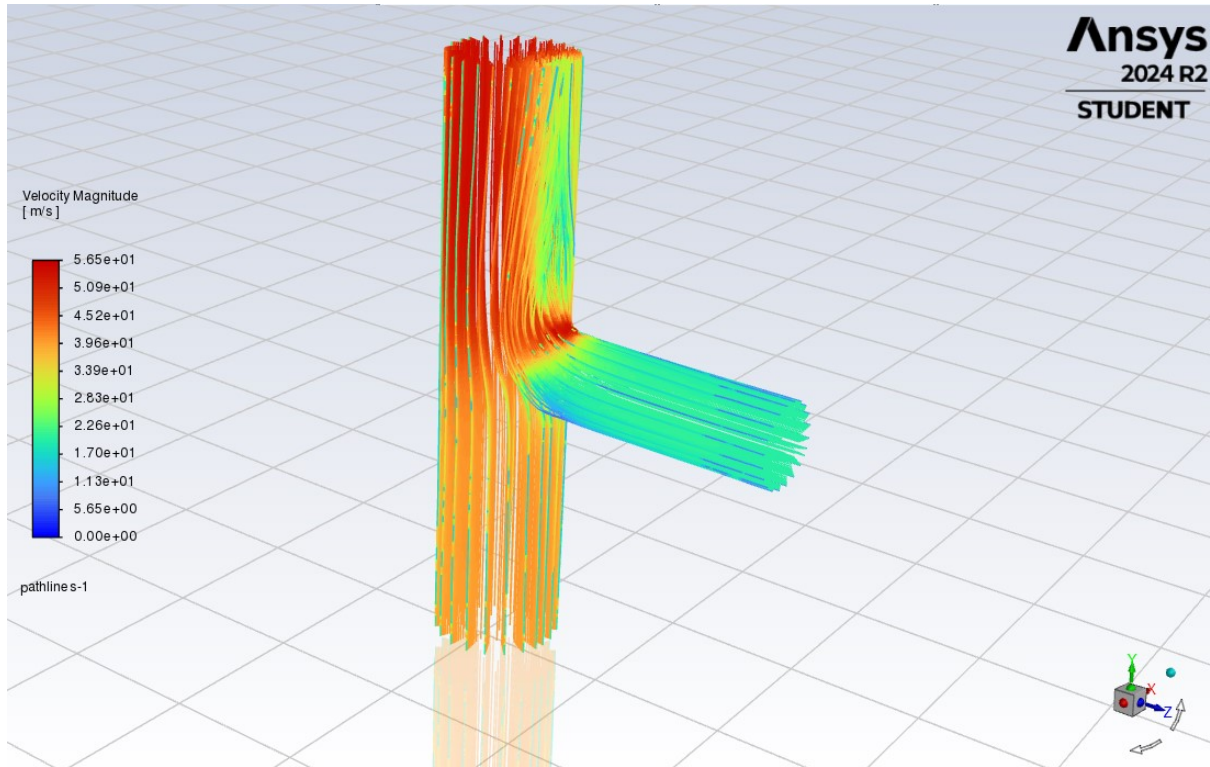
$$Error = \frac{|48.888889 - 48.740287|}{48.740287} \times 100 = 0.304\% \approx 0.3\%$$

This minimal error signifies strong agreement between the numerical simulation and theoretical analysis. The discrepancy can be attributed to numerical diffusion, turbulence modeling approximations, and finite resolution of the computational mesh.

3.3 Visualization and Analysis

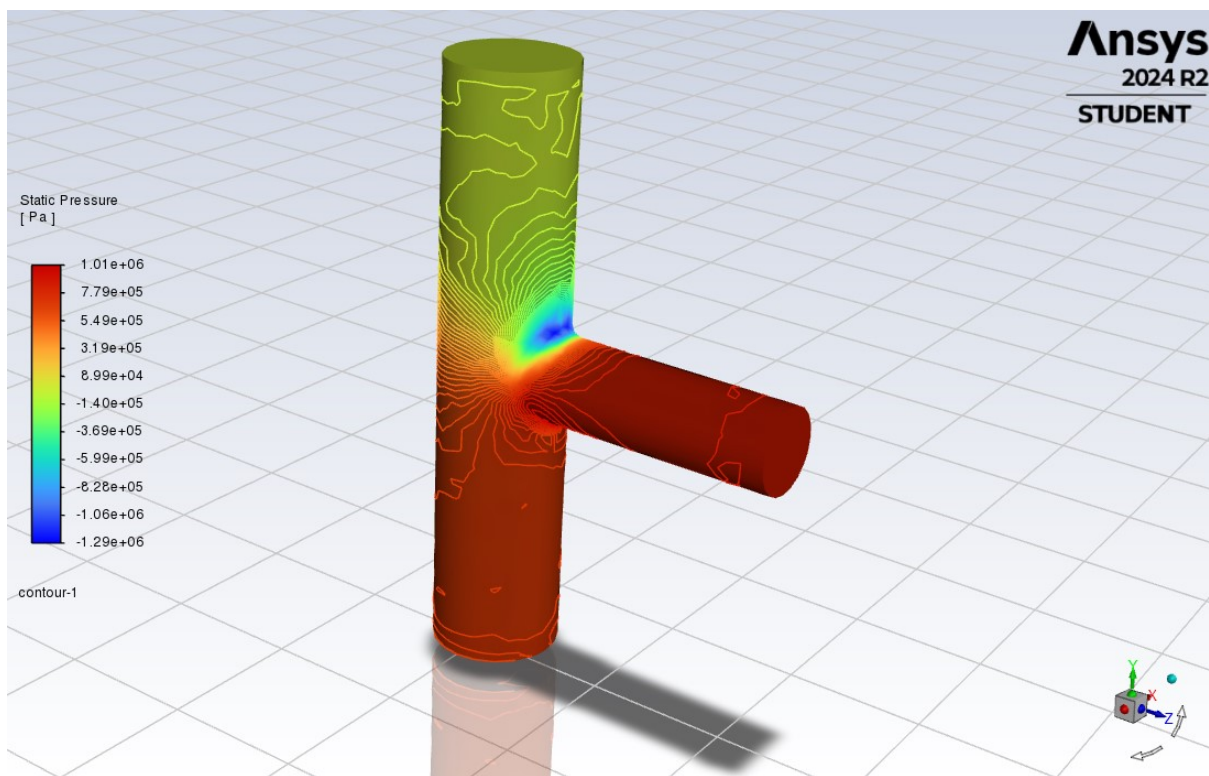
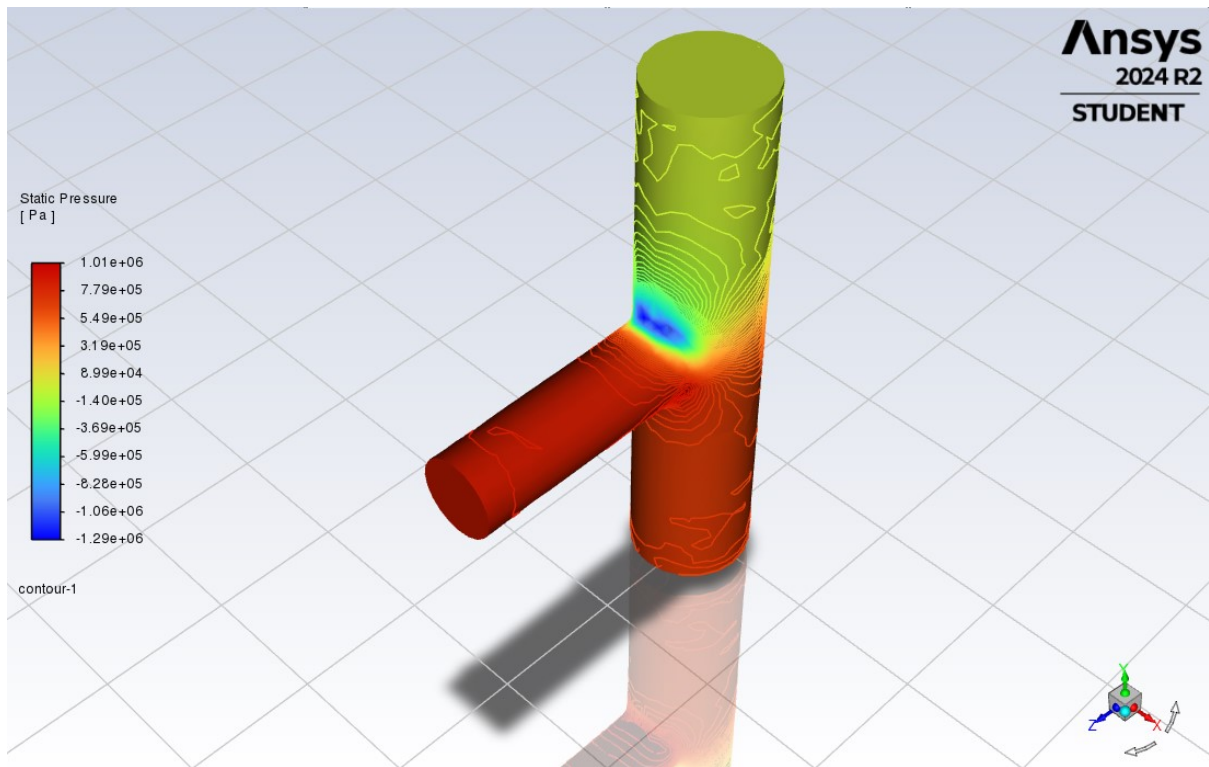
Velocity Field:

These pathlines revealed how the flow developed from both inlets and merged toward the outlet, highlighting regions of high velocity gradients and mixing near the T-junction.

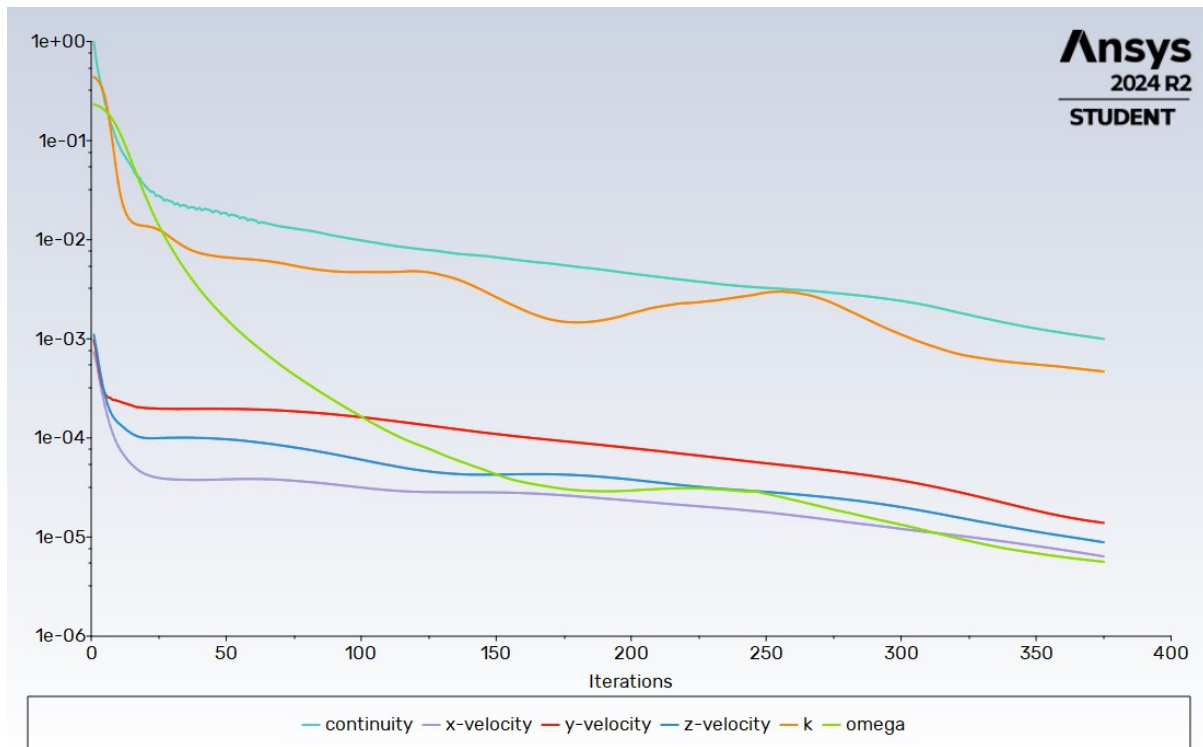


Pressure Distribution:

Pressure contours were plotted along the pipe centerline to identify regions of maximum and minimum pressure. The pressure contours, as shown in the simulation results, indicate significant variations near the T-junction where the flow from the secondary inlet interacts with the primary flow. The highest pressures occur upstream of the junction due to stagnation effects, while the lowest pressures are observed at the downstream side of the branch, reflecting acceleration and turbulence-induced pressure drops.



3.4 Convergence Observations



The convergence of the solution was analyzed by monitoring the residuals of the continuity, momentum, and turbulence equations. The residuals, as shown in the provided plot, exhibited the following behavior:

1. Continuity Residual:

The continuity residual successfully dropped below 10^{-3} , indicating adequate conservation of mass throughout the flow domain. This suggests that the solver was able to achieve an acceptable balance between the inlets and outlet, which is critical for incompressible flow simulations.

2. Momentum Residuals (X, Y, Z Components):

The momentum residuals for all velocity components (x, y, z) also satisfied the 10^{-3} threshold. The x-velocity residual demonstrated the smoothest decay, as expected due to the predominantly axial flow direction. The y and z-velocity residuals exhibited slightly slower convergence due to the influence of flow mixing and turbulence near the T-junction.

3. Turbulence Residuals (k and ω):

The turbulence quantities (k for kinetic energy and ω for specific dissipation rate) showed some initial fluctuations before stabilizing. These values eventually reduced to below 10^{-3} , aligning with the specified convergence criteria. The fluctuations observed early in the iterations can be attributed to the complex interaction between the two inlets and the transition to steady-state conditions.

Discussion of Residual Behavior

Early Iterations: Rapid decay in residuals was observed during the initial iterations as the solver adjusted to the boundary conditions and initialized the flow field.

Later Iterations: Residuals plateaued after approximately 250 iterations, indicating the solver was approaching a steady-state solution. Minor oscillations in the k and ω residuals were noted but did not significantly impact the stability or accuracy of the solution.

4. Conclusion

The simulated outlet velocity ~ 48.74 m/s exhibited a negligible error of 0.3% when compared to the theoretical prediction 48.89 m/s. This validates the robustness of the numerical model and simulation setup. The pressure distribution analysis revealed significant variations near the T-junction, with high-pressure zones upstream and low-pressure zones downstream, driven by the interaction of merging flows.

6. References

1. ANSYS Fluent User Guide, ANSYS Inc.
2. White, F.M. (2011). Fluid Mechanics, 7th Edition, McGraw-Hill Education.
3. Pope, S.B. (2000). Turbulent Flows, Cambridge University Press.
4. AE 301, aerodynamics course slides.